

Technical Report for the IMU Odometry Challenge

Replace with Your Submission Title

Kaggle team name: *exactly as it appears on the leaderboard*

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Abstract—Please read the latest notice from <https://superodometry.com/imuchallenge>

Provide a concise summary of your solution to the IMU Odometry Challenge. The abstract should briefly describe: (1) the core idea and overall approach; (2) the main model or system architecture; (3) the most important techniques used to improve performance; (4) the training data and inference strategy, when relevant; and (5) the final challenge results using the primary evaluation metrics.

Selection criteria. The Kaggle leaderboard is *not* the only criterion used by the challenge committee to determine the Top 10 teams: the quality of this technical report is also taken into account. The Top 10 teams will be invited to contribute to the forthcoming *IMU Foundation Model* white paper.

Format and deadline. Maximum **6 pages excluding references**, and **7 pages in total** including references, in this template. The **appendix does not count** towards either limit. Reports are due **23:59 US Eastern Time (EDT, UTC−4) on 23 September 2026**, three days after the leaderboard closes.

How to submit. The report is *not* uploaded on the website. It is submitted **inside the challenge Form**: fill the Form in as usual and attach this report at the end — that single submission is what we collect. The Form is linked from the **Announcements** section of <https://superodometry.com/imuchallenge>.

Please check that section regularly: all challenge news, rule clarifications and any change to these deadlines are posted there.

Where the numbers in this report come from. Every number in Tables III, IV and V must be produced by the **official per-sequence scoring service**:

https://huggingface.co/spaces/Tartan-IMU/imu_odometry_challenge_scoring

Upload your submission CSV and it returns ATE₂₀, AVE and RTE for all 89 test sequences, computed with the same code as the Kaggle leaderboard. **Five submissions per**

team per day. Your team name must match your **Kaggle team name exactly**; teams with no Kaggle submission are not scored.

I. INTRODUCTION

Briefly introduce your approach to the challenge. We recommend addressing the following questions:

- What are the main challenges of developing foundation models for IMU odometry?
- What motivated the design of your solution?
- What are the key components of your method?
- What distinguishes your method from others?
- What are the main contributions of your method?

II. METHOD

Describe the proposed method in sufficient detail to allow readers to understand, analyze, and reproduce the system.

A. Overall Architecture

- What is the overall system architecture of the proposed solution?
- What design choices were made specifically to improve cross-platform or out-of-distribution generalization?

Please include an architecture or pipeline figure.

Replace with method overview / architecture figure

Fig. 1. Overview of the proposed IMU odometry method.

B. Data Processing

- What is the data preprocessing process?
- What data augmentation (if any) is applied?
- Anything else?

Replace this footnote with funding or support acknowledgments, if applicable.

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C. Training Data

Clearly list **all datasets** used to develop or train the final submitted model. This includes official challenge data, external public datasets, synthetic data, self-collected data, pretrained checkpoints, and pseudo-labeled data.

TABLE I
DATASETS USED FOR MODEL DEVELOPMENT AND TRAINING.

Dataset	Type	Usage
Challenge train set	Official	Training / validation
External Dataset A	Public	Pretraining
Synthetic Dataset B	Synthetic	Augmentation

D. Training and Optimization

- What loss functions, optimizer, learning-rate schedule, batch size are used?
- How many epochs or optimization steps are used? What hardware is used for training? And what is the approximate training time?
- Is training performed in one stage or multiple stages?
- Is curriculum learning, fine-tuning, domain adaptation, or platform-specific specialization used?
- How is the final checkpoint selected?

E. Inference

- Describe the complete inference pipeline used to generate the final submission.
- What is the average inference latency or throughput?

F. Model Complexity and Runtime

- How many parameters does the final model contain?
- What hardware is used for inference?
- What is your inference speed? (Hz)
- Can the method operate online / in real time?
- What is the peak GPU/CPU memory requirement?

TABLE II
MODEL COMPLEXITY AND RUNTIME STATISTICS.

Property	Value
Number of parameters	–
Training hardware	–
Training time	–
Inference hardware	–
Inference speed (Hz)	–
Peak memory	–

III. DEVELOPMENT EXPERIMENTS

Summarize **what you actually tried and what happened** — the experimental record. Interpretation and lessons for the field belong in Section V, not here. Negative results are encouraged when they provide useful technical insight; please quantify gains rather than describing them qualitatively.

- What model architectures did you try?
- Which changes produced the largest gains, and **by how much** on which metric?

- Which approaches did not work as expected?
- Were some techniques beneficial for one platform but harmful for another?
- Which components improved generalization rather than only validation performance?
- Did more training data consistently help?
- Did model scale correlate with better performance?

IV. EXPERIMENTS

A. Official Challenge Performance

Report the performance of the final submitted method using the official challenge evaluation metrics, and include the leaderboard result or ranking.

All numbers in Tables III, IV and V must be produced by the official scoring service, https://huggingface.co/spaces/Tartan-IMU/imu_odometry_challenge_scoring, not computed locally. State the MD5 of the submission CSV they come from, so that the committee can reproduce them: md5 = [fill in].

TABLE III
OVERALL CHALLENGE PERFORMANCE ON THE **FINAL (PRIVATE) LEADERBOARD**. THE TARTANIMU SCORE IS $0.6 \text{ AVE}/0.7356 + 0.4 \text{ ATE}_{20}/3.1160$; IT IS DIMENSIONLESS AND *lower is better* (THE ALL-ZERO SUBMISSION SCORES EXACTLY 1.000). ATE_{20} IS THE 20 M-SEGMENT TRAJECTORY ERROR AFTER SE(3) ALIGNMENT. **RTE DOES NOT ENTER THE RANKING** AND IS REPORTED FOR ANALYSIS ONLY; IT IS DEFINED IN THE CAPTION OF TABLE V.

Method	Score ↓	ATE ₂₀ ↓	AVE ↓	RTE ↓
TartanIMU Baseline	–	–	–	–
Ours	–	–	–	–

B. Per-Platform Results

TABLE IV
PERFORMANCE BY PLATFORM. RTE IS DEFINED IN THE CAPTION OF TABLE V.

Platform	ATE ₂₀ ↓	AVE ↓	RTE ↓
Wheeled / Car	–	–	–
Handheld / Human	–	–	–
Legged / Quadruped	–	–	–
Aerial / Drone	–	–	–
Average	–	–	–

C. Per-Sequence Results

The full per-sequence table is reported in the appendix, Table V. Please discuss the trends here rather than repeating the numbers: which sequences are hardest, what they have in common, and whether the errors are concentrated in a few sequences or spread evenly.

D. Ablation Studies

Ablation studies are strongly encouraged for the most important components of the final method.

E. Generalization Analysis

- How well does the method generalize?
- Is there evidence of negative transfer between platforms?

F. Failure-Mode Analysis

- What are the most common failure cases?
- What limitations remain unresolved?

V. INSIGHTS AND LESSONS LEARNED

*This section is about **what your results mean for the field**, not about what you ran — the experimental record belongs in Section III. Please interpret those numbers rather than repeating them.*

- Based on your experiments, does a **single universal inertial model** appear feasible, or does platform specialization remain necessary? What evidence points either way?
- Where does the bottleneck currently sit — data, architecture, optimization, or the evaluation protocol itself?
- Which of your findings would you expect to **transfer** to platforms, sensors or motion regimes outside this benchmark, and which would not?
- What kinds of data gave the highest marginal benefit per hour collected?
- What did this challenge reveal about the current limitations of learning-based inertial odometry?
- Which research directions look most promising for future IMU odometry systems?

A. Feedback to the Organizers

This subsection is read directly by the challenge committee and feeds into the design of the next edition. Candid criticism is more useful to us than politeness.

- Were the rules, metrics, datasets and submission process clear and unambiguous? Which part cost you the most time to figure out?
- Did anything about the evaluation metric behave in a way you did not expect?
- What additional data, platforms, or evaluation settings would most improve the next edition?
- Was any part of the benchmark measuring something other than what it intended to?

VI. CONCLUSION

Summarize the overall report: the final solution and its main strengths, the most important remaining limitations, and the most promising future directions.

ACKNOWLEDGMENT

Please include the acknowledgment here.

APPENDIX

Add the appendix here.

A. Per-Sequence Results (Full Table)

*Report the metrics for **all 89 test sequences**. Upload your submission CSV to the official scoring service, https://huggingface.co/spaces/Tartan-IMU/imu_odometry_challenge_scoring, and it returns ATE, AVE and RTE for every sequence, ready to be copied into the table below (the service also offers the same table as a CSV download).*

B. Reproducibility and Submission Artifacts

List every artifact needed to reproduce the reported numbers: Kaggle team name and the submission that produced them (submission id and MD5 of the CSV); code repository and commit; trained checkpoint (link + MD5); configuration files; environment; random seeds, and whether the reported score is a single run or an average.

Compliance declaration (required). Every team must answer all five questions below explicitly. Reports that leave them unanswered cannot be considered for the Top 10.

- 1) Do the submitted predictions come from a **single model with one shared set of weights**? Inferring the embodiment inside the network is allowed and is the point of the benchmark; dispatching to separately trained per-platform experts is not.
- 2) Is any **platform classification, routing, or platform-specific specialization** used at inference? If yes, describe exactly where it sits in the pipeline.
- 3) Is any **ensembling** used — model ensembling, checkpoint averaging or soups, or test-time augmentation? If yes, state how many members and how they are combined.
- 4) How did you **avoid test-set leakage**, including any indirect use of test information for model or checkpoint selection?
- 5) If the development-time system differs from the submitted system, **what exactly changed** in the final submission?

C. Final Predictions

Provide a link to the final predictions on the challenge test set. This must be the same CSV that was submitted to Kaggle: exactly the columns below, one row per test window_id, 30,644 rows plus the header.

```
window_id,vx,vy,vz
1,0.0,0.0,0.0
2,0.0,0.0,0.0
...
```

v_x, v_y, v_z is the predicted mean **body-frame** velocity, in m/s, of that window. A window is 1.0 s = 200 frames at 200 Hz, non-overlapping: window k of a trajectory is `imu[k*200:(k+1)*200]`. Window ids are taken from `index/test_windows.csv`; every id must appear exactly once.

TABLE V

PER-SEQUENCE CHALLENGE RESULTS OVER THE 89 TEST SEQUENCES. ATE, AVE AND RTE ARE PRODUCED BY THE OFFICIAL PER-SEQUENCE SCORING SERVICE WHOSE URL IS GIVEN ON THE FIRST PAGE. **RTE** (RELATIVE TRAJECTORY ERROR, M) FOLLOWS THE DEFINITION OF STURM *et al.* [2] AND IS THE METRIC USED IN AIRIO: OVER SLIDING WINDOWS OF A FIXED DURATION Δt , IT IS THE RMSE OF THE DIFFERENCE BETWEEN THE PREDICTED AND THE GROUND-TRUTH RELATIVE DISPLACEMENT, $\text{RTE} = \sqrt{\frac{1}{n} \sum_i \|(\hat{\mathbf{p}}_{i+\Delta t} - \hat{\mathbf{p}}_i) - (\mathbf{p}_{i+\Delta t} - \mathbf{p}_i)\|^2}$. UNLIKE ATE IT MEASURES *local* DRIFT AND IS INSENSITIVE TO ACCUMULATED GLOBAL OFFSET. THE WINDOW LENGTH IS FIXED TO $\Delta t = 5$ S FOR ALL PLATFORMS: AT $\Delta t = 1$ S THE METRIC IS NEARLY A RESTATEMENT OF AVE, WHILE THE SHORTEST AERIAL SEQUENCES ARE ONLY 19 S LONG, WHICH BOUNDS Δt FROM ABOVE.

Seq. ID	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq. ID	ATE ₂₀ ↓	AVE ↓	RTE ↓	Seq. ID	ATE ₂₀ ↓	AVE ↓	RTE ↓
1	–	–	–	31	–	–	–	61	–	–	–
2	–	–	–	32	–	–	–	62	–	–	–
3	–	–	–	33	–	–	–	63	–	–	–
⋮				⋮				⋮			
30	–	–	–	60	–	–	–	89	–	–	–

D. Model and Inference Code

Please provide final trained model/checkpoint and inference script.

REFERENCES

- [1] S. Zhao, S. Zhou, R. Blanchard, Y. Qiu, W. Wang, and S. Scherer, “Tartan IMU: A Light Foundation Model for Inertial Positioning in Robotics,” in *Proc. IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 22520–22529.
- [2] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A Benchmark for the Evaluation of RGB-D SLAM Systems,” in *Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2012, pp. 573–580.
- [3] Replace with additional references as needed.